DEEP LEARNING		
Subject Code	**21AI62**	Credits:4
CIE:50	SEE:50	SEE:03hrs
Hours/Week:3hrs(Theory)		Total Hours:52Hrs
Prerequisite: Machine Learning		

Course Objectives: To enable the students to obtain the knowledge of DEEP LEARNING in the followingtopics.

- Understand complexity of Deep Learning algorithms and their limitations.
- Be capable of performing experiments in Deep Learning using real-world data

Modules	Teaching Hours
Module I **Introduction to Deep Learning**: Introduction to deep learning, Biological &artificial neurons ANN & its layer, Exploring activation functions, Forward propagation in ANN, How does ANN learn, Debugging gradient descent with gradient checking. Getting to Know **Tensor Flow:** Introduction to Tensor Flow, Understanding computational graphs and sessions, Variables, constants and placeholders, Introducing Tensor Board , Handwritten digit classification using Tensor Flow , Introducing eager execution,MathoperationsinTensorFlow,TensorFlow2.0andKeras,Kerasor Tensor Flow	**11 Hours**
Module II **Introduction to RNN:** Generating Song Lyrics Using RNN, Introducing RNNs Generatingsong lyrics using RNNs, Different types of RNN architectures. **Improvements to the RNN**: Improvements to the RNN, LSTM to the rescue, Gated recurrent units, Bi directional RNN, Going deep with deep RNN Language translation using the seq2seq model.	**10 Hours**
Module III **Demystifying Convolutional Networks**: Demystifying Convolutional Networks, Introduction to CNNs , The architecture of CNNs ,The math behind CNNs , Implementing a CNN in Tensor Flow, CNN architectures, Capsule networks, Building Capsule networks in Tensor Flow. Case study	**10 Hours**

Module IV	
Learning Text Representations: Learning Text Representations ,Understanding the word2vec model ,Building the word2vec model using gensim, Visualizing word embeddingsinTensorBoard,Doc2vecUnderstanding,skip-thoughtsalgorithm ,Quick thoughts for sentence embeddings.	**10 Hours**
Module V	
Generating Images Using GANs: Generating Images Using GANs, Differences between discriminative and generative models. DCGAN –Adding convolution to a GAN,Deconvolution generator, convolutional discriminator. **Learning More about GANs**: Conditional GAN, Loss Function of CGAN, GeneratingspecificdigitsusingCGAN,UnderstandingInfoGAN,ExploringMutual Information, Architecture of Info GAN, Translating images using Cycle GAN, Role Of generators, Role of discriminators, Loss Function, Cycle Consistency Loss, Stack GAN,Architecture of Stack GANs. Introduction to auto encoder.	**11 Hours**

Question paper pattern:

1. The question paper wills have TEN questions.
2. There will be TWO questions in each module, covering all the topics.
3. The student need to answer FIVE full questions, selecting ONE full question from each module.

Textbooks:

Cosma Rohilla Shalizi, Advanced Data Analysis from an Elementary Point of View,2020.

Course outcomes: On completion of the course ,the student twill have the ability to:

Course Code	CO#	Course Outcome(CO)
	CO1	Understand the concepts o f Deep Learning, Tensor Flow, its main functions, operations and the execution pipeline
	CO2	Understand Recurrent Neural Networks(RNN),Implement different architectures of RNN in Tensor flow
	CO3	Learn convolutional neural networks, Implement different architectures of CNN in Tensor flow
	CO4	Demonstrate Text Representations and Build the world have model using genesis and interpret the results.
	CO5	Build different architectures of GANS in Tensor flow